```
import pandas as pd
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from sklearn.preprocessing import MinMaxScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from google.colab import drive, files
```

```
from google.colab import drive
drive.mount('/content/drive', force_remount = True)
```

```
Mounted at /content/drive
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
file_path = '/content/drive/My Drive/Upgrad MS Datasets/Combined_Cricket_Data.csv'

# 3. Read the 4th sheet (index 3)
df_combined = pd.read_csv(file_path)
```

```
df_combined
```

	Player	Category	Country	Role	Runs Scored	Batting Strike Rate	Batting Average	No of Sixes	No of Fours	Balls Faced	...	Fours Conceded	Dot Balls Bowled	Maidens Bowled	Age Pla
0	Hardik Pandya	International	India	Fast-All Rounder	2757.0	147.00	28.34	149.0	207.0	1875.0	...	222.0	582.0	1.0	
1	Ravindra Jadeja	International	India	Spin-All Rounder	2356.0	134.30	28.10	82.0	175.0	1754.0	...	265.0	1116.0	2.0	
2	Shivam Dube	International	India	Fast-All Rounder	903.0	149.80	30.48	57.0	56.0	603.0	...	53.0	159.0	0.0	
3	Nitish Kumar Reddy	International	India	Fast-All Rounder	485.0	132.90	28.52	25.0	31.0	365.0	...	16.0	39.0	0.0	
4	Axar Patel	International	India	Spin-All Rounder	1910.0	134.20	22.02	94.0	129.0	1423.0	...	259.0	1132.0	2.0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
1873	Player_346	International	Australia	Batsman	866.0	197.16	34.68	39.0	94.0	344.0	...	NaN	NaN	NaN	
1874	Player_347	International	India	Batsman	1880.0	132.45	34.96	75.0	199.0	988.0	...	NaN	NaN	NaN	
1875	Player_348	International	West Indies	Batsman	1151.0	146.66	24.67	55.0	71.0	798.0	...	NaN	NaN	NaN	
1876	Player_349	International	England	Batsman	7088.0	147.57	41.70	124.0	791.0	5123.0	...	NaN	NaN	NaN	
1877	Player_350	Domestic	India	Batsman	645.0	132.40	25.07	20.0	52.0	311.0	...	NaN	NaN	NaN	

1878 rows × 31 columns

```
df_combined['Last Auction Price Sold (in Crs)'].isnull().sum()
```

```
np.int64(0)
```

```
import numpy as np

# Define the column for auction price
auction_price_col = 'Last Auction Price Sold (in Crs)'

# Ensure the column exists in df_scrambled
if auction_price_col in df_combined.columns:
    # Define the new thresholds based on user request
    low_threshold_value = 0.5  # 1 Crore
```

```
    high_threshold_value = 7 # 7 Crores

    # Define bins and labels based on the new thresholds
    # Order: (-inf, 1] -> Low, (1, 7] -> Mid, (7, inf) -> High
    bins = [-np.inf, low_threshold_value, high_threshold_value, np.inf]
    labels = ['Low Value', 'Mid Value', 'High Value']

    # Create the new categorical column 'Player_Value_Category'
    df_combined['Player_Value_Category'] = pd.cut(df_combined[auction_price_col], bins=bins, labels=labels, include_lowest=True

    print(f"Player Value Categories created based on '{auction_price_col}' with new thresholds:")
    display(df_combined['Player_Value_Category'].value_counts())

    # Display the first few rows with the new column
    print("\nFirst 5 rows of df_scrambled with 'Player_Value_Category':")
    display(df_combined[[auction_price_col, 'Player_Value_Category']].head())
else:
    print(f"Error: '{auction_price_col}' column not found in df_scrambled. Please ensure the column name is correct.")
```

```
Player Value Categories created based on 'Last Auction Price Sold (in Crs)' with new thresholds:
```

Player_Value_Category	count
Mid Value	1095
Low Value	488
High Value	295

dtype: int64

```
First 5 rows of df_scrambled with 'Player_Value_Category':
```

	Last Auction Price Sold (in Crs)	Player_Value_Category
0	16.35	High Value
1	18.00	High Value
2	12.00	High Value
3	6.00	Mid Value
4	16.50	High Value

```
df_combined ['Player_Value_Category']
```

	Player_Value_Category
0	High Value
1	High Value
2	High Value
3	Mid Value
4	High Value
...	...
1873	High Value
1874	Mid Value
1875	Mid Value
1876	Mid Value
1877	Mid Value

1878 rows × 1 columns

dtype: category

Preprocessing and Scrambling Data

To 'scramble' the data for potential machine learning tasks, we will perform the following steps on the `df_combined` DataFrame, which now includes the `Player_Value_Category`:

1. **Anonymize Player Names**: Replace player names with generic IDs.
2. **One-Hot Encode Categorical Features**: Convert categorical columns into a numerical format suitable for models.
3. **Scale Numerical Features**: Normalize numerical features to a standard range.
4. **Shuffle the dataset**: Randomly reorder the rows of the DataFrame.

```
# 1. Anonymize Player Names
df_processed = df_combined.copy()
df_processed['Player_ID'] = range(1, len(df_processed) + 1)
df_processed = df_processed.drop('Player', axis=1)

# Define categorical and numerical features for preprocessing
categorical_features = ['Category', 'Country', 'Role', 'Capped/Uncapped', 'Player_Value_Category']

# Dynamically get numerical features, excluding Player_ID which is an identifier
numerical_features = df_processed.select_dtypes(include=np.number).columns.tolist()
if 'Player_ID' in numerical_features:
    numerical_features.remove('Player_ID')

# 2. One-Hot Encode Categorical Features and 3. Scale Numerical Features
# Using ColumnTransformer for robust preprocessing
preprocessor = ColumnTransformer(
    transformers=[
        ('num', MinMaxScaler(), numerical_features),
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features)
    ],
    remainder='passthrough' # Keep other columns (like Player_ID) as they are
)

df_transformed = preprocessor.fit_transform(df_processed)

# Get feature names after one-hot encoding
categorical_feature_names = preprocessor.named_transformers_['cat'].get_feature_names_out(categorical_features)

# Get original columns that were not transformed (e.g., Player_ID which is handled by remainder)
original_cols_not_transformed = [col for col in df_processed.columns if col not in numerical_features + categorical_features]

# Create the final column names list in the correct order
final_column_names = numerical_features + list(categorical_feature_names) + original_cols_not_transformed

# Create a DataFrame from the transformed data using the correct column names
df_scrambled = pd.DataFrame(df_transformed, columns=final_column_names)

# 4. Shuffle the dataset
df_scrambled = df_scrambled.sample(frac=1, random_state=42).reset_index(drop=True)

print("\nFirst 5 rows of the preprocessed and scrambled (anonymized, encoded, scaled, shuffled) DataFrame:")
display(df_scrambled.head())
print("\nShape of the preprocessed and scrambled DataFrame:", df_scrambled.shape)
```

```
First 5 rows of the preprocessed and scrambled (anonymized, encoded, scaled, shuffled) DataFrame:
```

	Runs Scored	Batting Strike Rate	Batting Average	No of Sixes	No of Fours	Balls Faced	Total career dot balls played	No of Centuries Scored	No of Half Centuries Scored	Wickets Taken	...	Role_Spin All Rounder	Role_Spin Bowler	R
0	NaN	0.066734	0.247088	0.055238	0.024263	0.048411	0.060671	0.023256	0.034335	NaN	...	0.0	0.0	
1	0.023316	0.092906	0.201024	0.064127	0.017092	0.027101	0.034389	0.011628	0.023605	NaN	...	0.0	0.0	
2	NaN	0.068498	0.187152	0.024762	0.011788	0.021700	0.023551	0.000000	0.017167	NaN	...	0.0	0.0	
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.083838	...	0.0	0.0	
4	0.015564	0.066587	0.215043	0.033651	0.007466	0.020508	0.023466	0.000000	0.017167	0.045808	...	0.0	0.0	

5 rows × 56 columns

```
Shape of the preprocessed and scrambled DataFrame: (1878, 56)
```

```
# Define features (X) and target (y)
X = df_resampled.drop('Player_Value_Category', axis=1)
y = df_resampled['Player_Value_Category']

# Split into training (65%) and temporary (35%) sets
```

```
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.25, random_state=42, stratify=y)

# Split temporary set into testing (25% of original) and validation (10% of original)
# Calculate the test_size relative to the temporary set (0.10/0.35) for validation
X_test, X_val, y_test, y_val = train_test_split(X_temp, y_temp, test_size=(0.10/0.25), random_state=42, stratify=y_temp)

print(f"\nShape of training set: {X_train.shape}")
print(f"Shape of testing set: {X_test.shape}")
print(f"Shape of validation set: {X_val.shape}")

print("\nDistribution of Player_Value_Category in training set:")
display(y_train.value_counts(normalize=False))
print("\nDistribution of Player_Value_Category in testing set:")
display(y_test.value_counts(normalize=False))
print("\nDistribution of Player_Value_Category in validation set:")
display(y_val.value_counts(normalize=False))
```

```
Shape of training set: (675, 53)
Shape of testing set: (135, 53)
Shape of validation set: (90, 53)

Distribution of Player_Value_Category in training set:
```

Player_Value_Category	count
High Value	375
Low Value	150
Mid Value	150

dtype: int64

```
Distribution of Player_Value_Category in testing set:
```

Player_Value_Category	count
High Value	75
Low Value	30
Mid Value	30

dtype: int64

```
Distribution of Player_Value_Category in validation set:
```

Player_Value_Category	count
High Value	50
Low Value	20
Mid Value	20

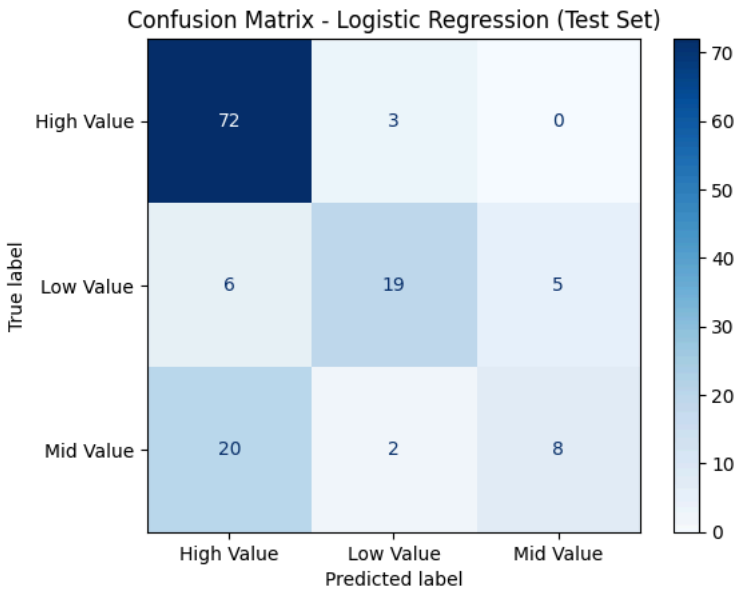
dtype: int64

```
# Impute NaN values in X_train, X_test, X_val with 0
X_train = X_train.fillna(0)
X_test = X_test.fillna(0)
X_val = X_val.fillna(0)

print("Missing values in X_train after imputation:")
display(X_train.isnull().sum().sum())

print("Missing values in X_test after imputation:")
display(X_test.isnull().sum().sum())

print("Missing values in X_val after imputation:")
display(X_val.isnull().sum().sum())
```

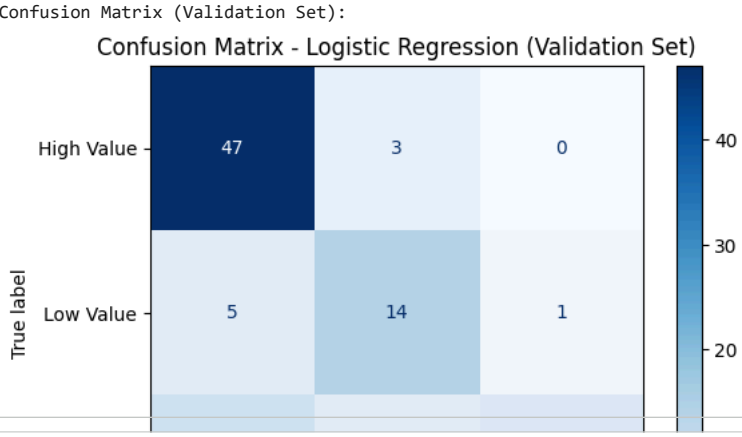
```
Missing values in X_train after imputation:
np.int64(0)
Missing values in X_test after imputation:
np.int64(0)
Missing values in X_val after imputation:
np.int64(0)
```

```
from sklearn.linear_model import LogisticRegression

# Initialize and train the Logistic Regression model
log_reg_model = LogisticRegression(max_iter=1000, random_state=42, solver='liblinear')
log_reg_model.fit(X_train, y_train)

# Evaluate the model
evaluate_model(log_reg_model, X_test, y_test, X_val, y_val, "Logistic Regression")
```

```
--- Evaluating Logistic Regression ---

Test Set Metrics for Logistic Regression:
Accuracy: 0.7333
Precision: 0.7208
Recall: 0.7333
Confusion Matrix (Test Set):
```



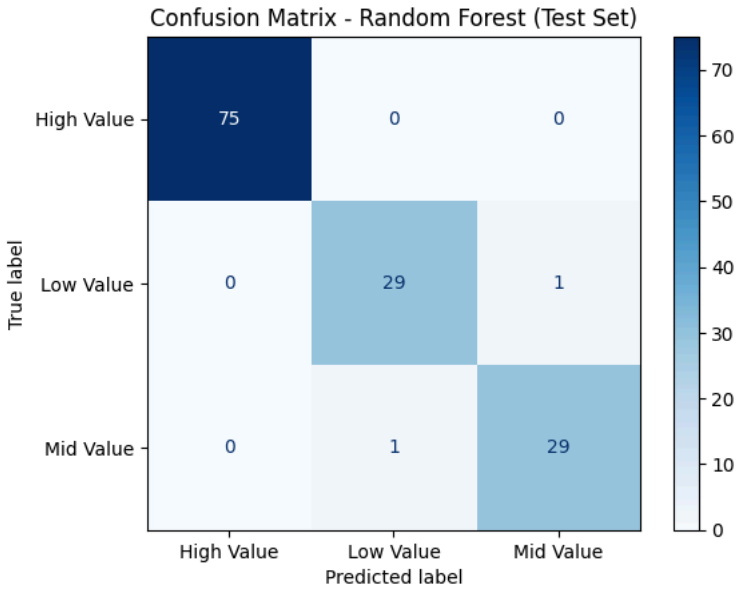

```
Validation Set Metrics for Logistic Regression:
Accuracy: 0.7444
Precision: 0.7599
Recall: 0.7444
Confusion Matrix (Validation Set):
```



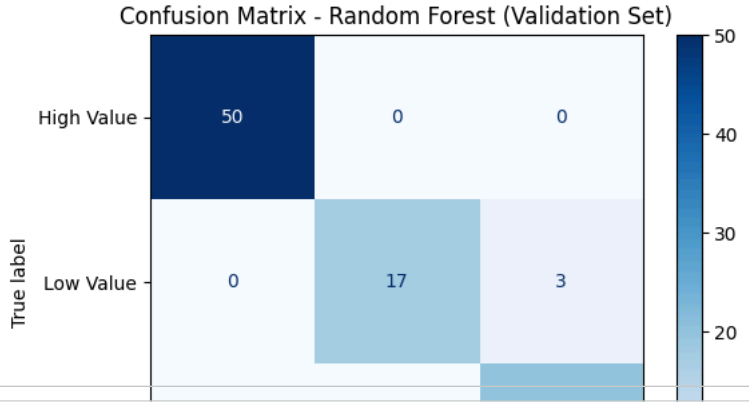

```
from sklearn.ensemble import RandomForestClassifier

# Initialize and train the Random Forest model
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)
```

```
# Evaluate the model
evaluate_model(rf_model, X_test, y_test, X_val, y_val, "Random Forest")
```

```
--- Evaluating Random Forest ---

Test Set Metrics for Random Forest:
Accuracy: 0.9852
Precision: 0.9852
Recall: 0.9852
Confusion Matrix (Test Set):
```

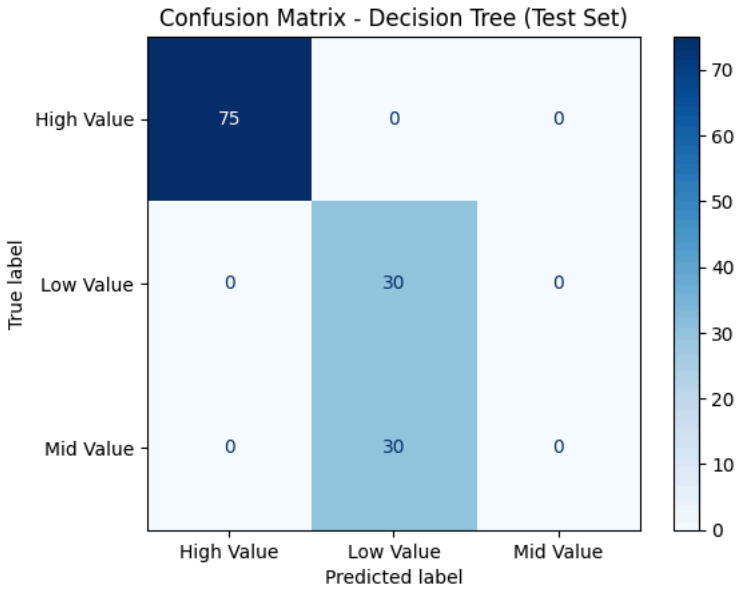



```
Validation Set Metrics for Random Forest:
Accuracy: 0.9667
Precision: 0.9710
Recall: 0.9667
Confusion Matrix (Validation Set):
```

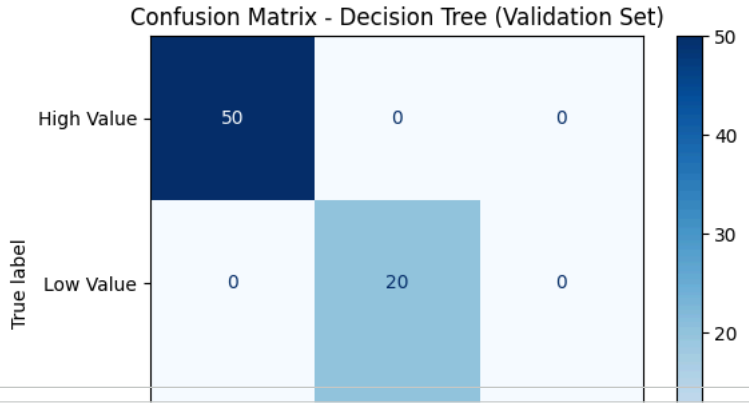



```
from sklearn.tree import DecisionTreeClassifier

# Initialize and train the Decision Tree model
dt_model = DecisionTreeClassifier(random_state=42,max_depth=1)
dt_model.fit(X_train, y_train)

# Evaluate the model
evaluate_model(dt_model, X_test, y_test, X_val, y_val, "Decision Tree")
```

```
--- Evaluating Decision Tree ---

Test Set Metrics for Decision Tree:
Accuracy: 0.7778
Precision: 0.6667
Recall: 0.7778
Confusion Matrix (Test Set):
```

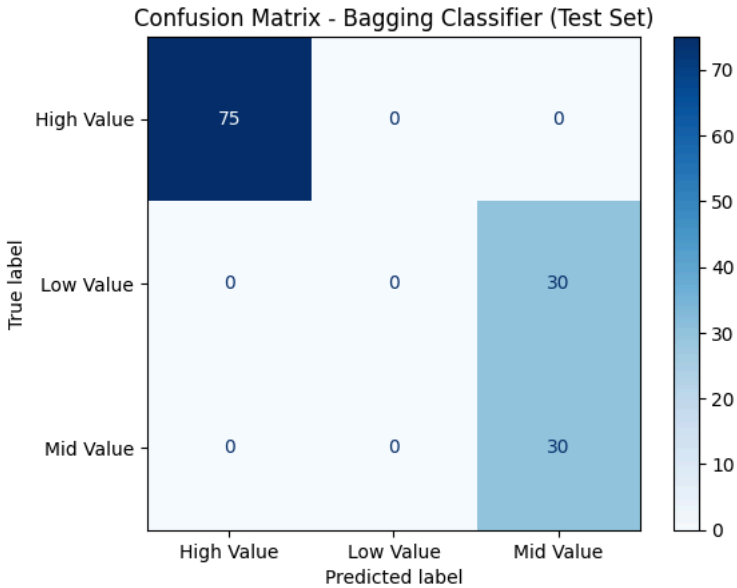



```
Validation Set Metrics for Decision Tree:
Accuracy: 0.7778
Precision: 0.6667
Recall: 0.7778
Confusion Matrix (Validation Set):
```

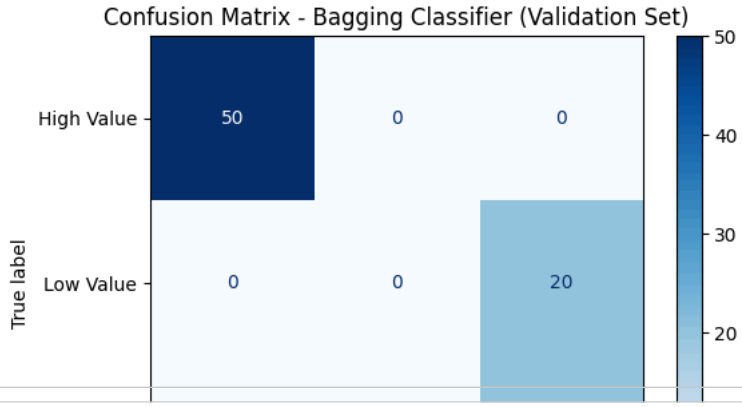



```
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier

# Initialize and train the Bagging Classifier model
bagging_model = BaggingClassifier(estimator=DecisionTreeClassifier(random_state=42,max_depth=1), random_state=42)
bagging_model.fit(X_train, y_train)

# Evaluate the model
evaluate_model(bagging_model, X_test, y_test, X_val, y_val, "Bagging Classifier")
```

```
--- Evaluating Bagging Classifier ---

Test Set Metrics for Bagging Classifier:
Accuracy: 0.7778
Precision: 0.6667
Recall: 0.7778
Confusion Matrix (Test Set):
```

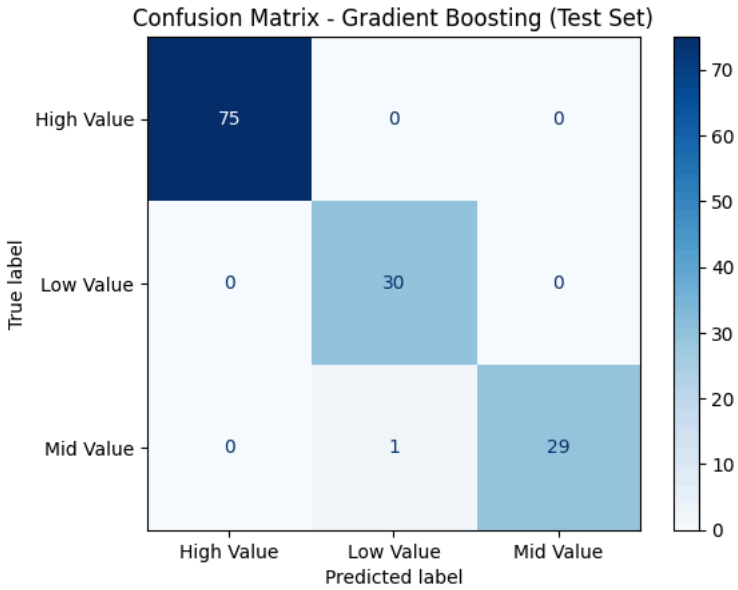



```
Validation Set Metrics for Bagging Classifier:
Accuracy: 0.7778
Precision: 0.6667
Recall: 0.7778
Confusion Matrix (Validation Set):
```

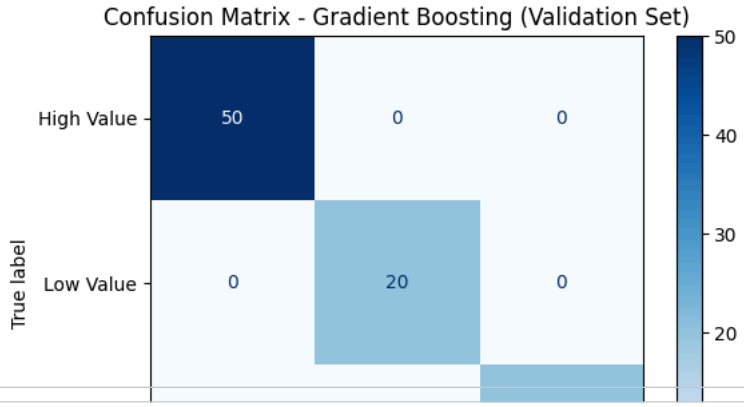



```
from sklearn.ensemble import GradientBoostingClassifier

# Initialize and train the Gradient Boosting model
gb_model = GradientBoostingClassifier(random_state=42)
gb_model.fit(X_train, y_train)

# Evaluate the model
evaluate_model(gb_model, X_test, y_test, X_val, y_val, "Gradient Boosting")
```

```
--- Evaluating Gradient Boosting ---

Test Set Metrics for Gradient Boosting:
Accuracy: 0.9926
Precision: 0.9928
Recall: 0.9926
Confusion Matrix (Test Set):
```



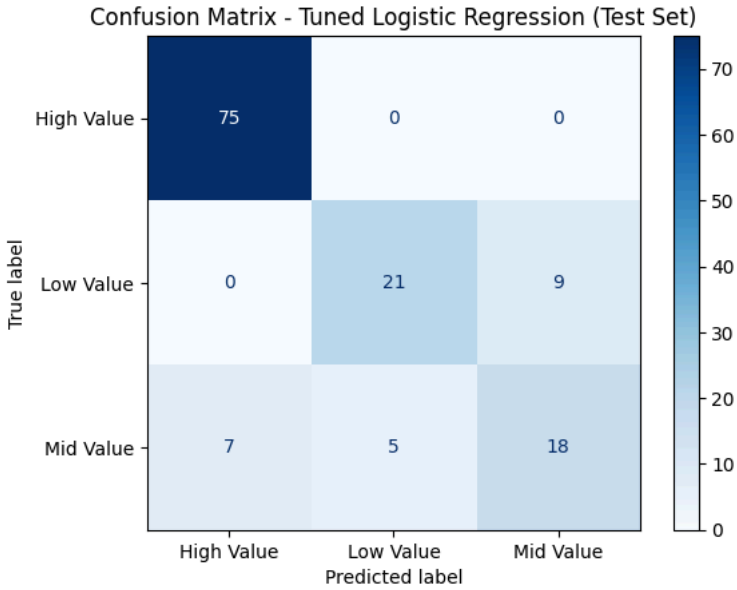

```
Validation Set Metrics for Gradient Boosting:
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
Confusion Matrix (Validation Set):
```



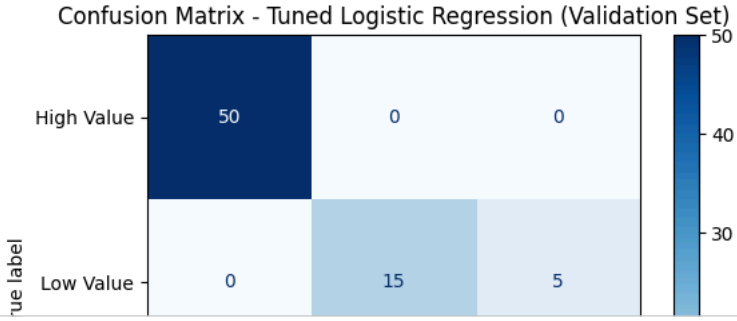

```
from sklearn.model_selection import GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, BaggingClassifier, GradientBoostingClassifier
from sklearn.tree import DecisionTreeClassifier

# --- Logistic Regression Hyperparameter Tuning ---
print("\n--- Hyperparameter Tuning for Logistic Regression ---")
param_grid_lr = {
    'C': [0.01, 0.1, 1, 10, 100],
    'solver': ['liblinear', 'saga']
}
grid_search_lr = GridSearchCV(LogisticRegression(max_iter=1000, random_state=42), param_grid_lr, cv=3, scoring='accuracy', n_jo
grid_search_lr.fit(X_train, y_train)

best_lr_model = grid_search_lr.best_estimator_
print(f"Best parameters for Logistic Regression: {grid_search_lr.best_params_}")
evaluate_model(best_lr_model, X_test, y_test, X_val, y_val, "Tuned Logistic Regression")

# --- Random Forest Hyperparameter Tuning ---
print("\n--- Hyperparameter Tuning for Random Forest ---")
param_grid_rf = {
    'n_estimators': [50, 100, 200],
    'max_depth': [None, 10, 20],
```

```
    'min_samples_split': [2, 5]
}
grid_search_rf = GridSearchCV(RandomForestClassifier(random_state=42), param_grid_rf, cv=3, scoring='accuracy', n_jobs=-1)
grid_search_rf.fit(X_train, y_train)

best_rf_model = grid_search_rf.best_estimator_
print(f"Best parameters for Random Forest: {grid_search_rf.best_params_}")
evaluate_model(best_rf_model, X_test, y_test, X_val, y_val, "Tuned Random Forest")

# --- Decision Tree Hyperparameter Tuning ---
print("\n--- Hyperparameter Tuning for Decision Tree ---")
param_grid_dt = {
    'max_depth': [None, 10, 20, 30],
    'min_samples_leaf': [1, 5, 10]
}
grid_search_dt = GridSearchCV(DecisionTreeClassifier(random_state=42), param_grid_dt, cv=3, scoring='accuracy', n_jobs=-1)
grid_search_dt.fit(X_train, y_train)

best_dt_model = grid_search_dt.best_estimator_
print(f"Best parameters for Decision Tree: {grid_search_dt.best_params_}")
evaluate_model(best_dt_model, X_test, y_test, X_val, y_val, "Tuned Decision Tree")

# --- Bagging Classifier Hyperparameter Tuning ---
print("\n--- Hyperparameter Tuning for Bagging Classifier ---")
param_grid_bag = {
    'n_estimators': [10, 50, 100],
    'max_samples': [0.8, 1.0]
}
grid_search_bag = GridSearchCV(BaggingClassifier(estimator=DecisionTreeClassifier(random_state=42), random_state=42), param_gri
grid_search_bag.fit(X_train, y_train)

best_bag_model = grid_search_bag.best_estimator_
print(f"Best parameters for Bagging Classifier: {grid_search_bag.best_params_}")
evaluate_model(best_bag_model, X_test, y_test, X_val, y_val, "Tuned Bagging Classifier")

# --- Gradient Boosting Hyperparameter Tuning ---
print("\n--- Hyperparameter Tuning for Gradient Boosting ---")
param_grid_gb = {
    'n_estimators': [50, 100, 200],
    'learning_rate': [0.01, 0.1, 0.2],
    'max_depth': [3, 5]
}
grid_search_gb = GridSearchCV(GradientBoostingClassifier(random_state=42), param_grid_gb, cv=3, scoring='accuracy', n_jobs=-1)
grid_search_gb.fit(X_train, y_train)

best_gb_model = grid_search_gb.best_estimator_
print(f"Best parameters for Gradient Boosting: {grid_search_gb.best_params_}")
evaluate_model(best_gb_model, X_test, y_test, X_val, y_val, "Tuned Gradient Boosting")
```

```
--- Hyperparameter Tuning for Logistic Regression ---
Best parameters for Logistic Regression: {'C': 100, 'solver': 'liblinear'}

--- Evaluating Tuned Logistic Regression ---

Test Set Metrics for Tuned Logistic Regression:
Accuracy: 0.8444
Precision: 0.8358
Recall: 0.8444
Confusion Matrix (Test Set):
```



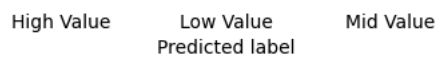

```
Validation Set Metrics for Tuned Logistic Regression:
Accuracy: 0.8556
Precision: 0.8498
Recall: 0.8556
Confusion Matrix (Validation Set):
```



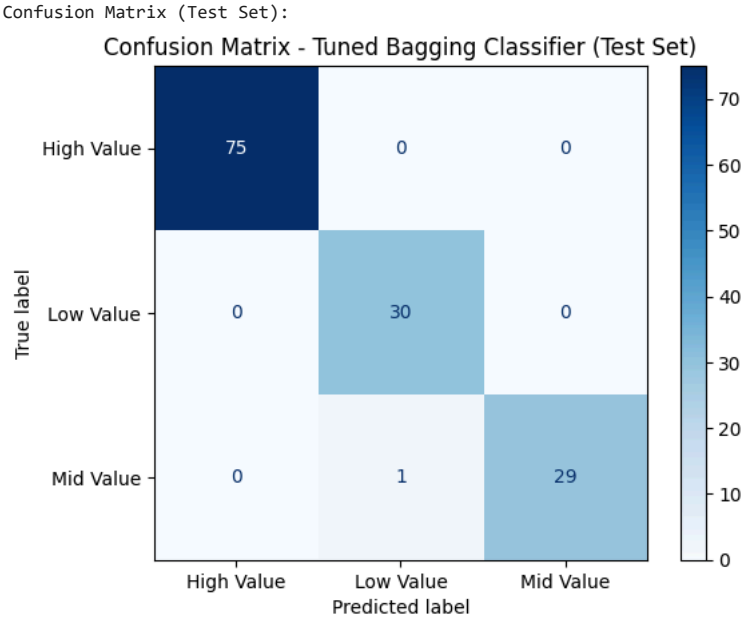

```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix, ConfusionMatrixDisplay # Added imp

# 1. Convert data to PyTorch tensors
# First, encode the target labels (y) to numerical values if they are strings
label_encoder = LabelEncoder()
y_train_encoded = label_encoder.fit_transform(y_train)
y_test_encoded = label_encoder.transform(y_test)
y_val_encoded = label_encoder.transform(y_val)

X_train_tensor = torch.tensor(X_train.values.astype(np.float32))
y_train_tensor = torch.tensor(y_train_encoded, dtype=torch.long)
X_test_tensor = torch.tensor(X_test.values.astype(np.float32))
y_test_tensor = torch.tensor(y_test_encoded, dtype=torch.long)
X_val_tensor = torch.tensor(X_val.values.astype(np.float32))
y_val_tensor = torch.tensor(y_val_encoded, dtype=torch.long)

# Create DataLoader for batch processing
train_dataset = TensorDataset(X_train_tensor, y_train_tensor)
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
```

```
test_dataset = TensorDataset(X_test_tensor, y_test_tensor)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)

val_dataset = TensorDataset(X_val_tensor, y_val_tensor)
val_loader = DataLoader(val_dataset, batch_size=32, shuffle=False)

# 2. Define the CNN-like model architecture
class TabularCNN(nn.Module):
    def __init__(self, input_dim, num_classes):
        super(TabularCNN, self).__init__()
        # Using 1D convolution as a feature extractor, then fully connected layers
        # Reshape input to be (batch_size, channels, features) for 1D conv
        # Here, we treat each feature as a 'channel' for simplicity or could add a dummy channel dim
        self.conv1 = nn.Conv1d(in_channels=1, out_channels=16, kernel_size=3, padding=1)
        self.pool = nn.MaxPool1d(kernel_size=2)
        self.relu = nn.ReLU()

        # Calculate the output size from convolutional layers
        # Input shape: (batch_size, input_dim)
        # After unsqueeze: (batch_size, 1, input_dim)
        # After conv1: (batch_size, 16, input_dim)
        # After pool: (batch_size, 16, input_dim // 2) (assuming input_dim is even)
        # Flatten for fully connected layer
        # We need to compute this dynamically, or ensure the dimensions align

        # A simpler approach for tabular data often bypasses conv layers for direct FC layers
        # Let's use a fully connected network to represent 'CNN-like' processing for tabular data
        # as direct 1D CNN for raw tabular features can be tricky without careful reshaping

        self.fc1 = nn.Linear(input_dim, 128)  # First fully connected layer
        self.fc2 = nn.Linear(128, 64)
        self.fc3 = nn.Linear(64, num_classes) # Output layer
        self.dropout = nn.Dropout(0.5)

    def forward(self, x):
        # For tabular data, direct fully connected layers are more common.
        # If using 1D Conv, reshape x: x = x.unsqueeze(1) for (batch_size, 1, features)
        # Here, we'll proceed with fully connected layers as a robust approach for tabular data.
        x = self.relu(self.fc1(x))
        x = self.dropout(x)
        x = self.relu(self.fc2(x))
        x = self.dropout(x)
        x = self.fc3(x)
        return x

# Determine input dimension and number of classes
input_dim = X_train.shape[1]
num_classes = len(label_encoder.classes_)

cnn_model = TabularCNN(input_dim, num_classes)

# 3. Train the model
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(cnn_model.parameters(), lr=0.001)

num_epochs = 50

print("\n--- Training CNN Model ---")
for epoch in range(num_epochs):
    cnn_model.train()
    running_loss = 0.0
    for inputs, labels in train_loader:
        optimizer.zero_grad()
        outputs = cnn_model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item()
    # print(f"Epoch {epoch+1}/{num_epochs}, Loss: {running_loss/len(train_loader):.4f}")

print("CNN Model Training Complete.")

# 4. Evaluate the CNN model (custom evaluation as evaluate_model is for sklearn)
def evaluate_cnn_model(model, data_loader, label_encoder, model_name, set_name):
    model.eval() # Set the model to evaluation mode
    all_labels = []
```

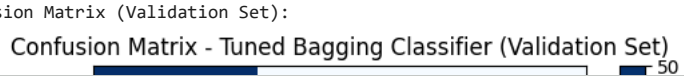
```
    all_predictions = []

    with torch.no_grad():
        for inputs, labels in data_loader:
            outputs = model(inputs)
            _, predicted = torch.max(outputs.data, 1)
            all_labels.extend(labels.cpu().numpy())
            all_predictions.extend(predicted.cpu().numpy())

    # Convert encoded labels back to original string labels for metrics calculation
    original_labels = label_encoder.inverse_transform(all_labels)
    original_predictions = label_encoder.inverse_transform(all_predictions)

    accuracy = accuracy_score(original_labels, original_predictions)
    precision = precision_score(original_labels, original_predictions, average='weighted', zero_division=0)
    recall = recall_score(original_labels, original_predictions, average='weighted', zero_division=0)
    cm = confusion_matrix(original_labels, original_predictions, labels=label_encoder.classes_)

    print(f"\n{set_name} Set Metrics for {model_name}:")
    print(f"Accuracy: {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall: {recall:.4f}")
    print("Confusion Matrix (Test Set):")
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=label_encoder.classes_)
    disp.plot(cmap=plt.cm.Blues)
    plt.title(f'Confusion Matrix - {model_name} (Test Set)')
    plt.show()

# Evaluate on Test Set
evaluate_cnn_model(cnn_model, test_loader, label_encoder, "CNN Classifier", "Test")

# Evaluate on Validation Set
evaluate_cnn_model(cnn_model, val_loader, label_encoder, "CNN Classifier", "Validation")
```



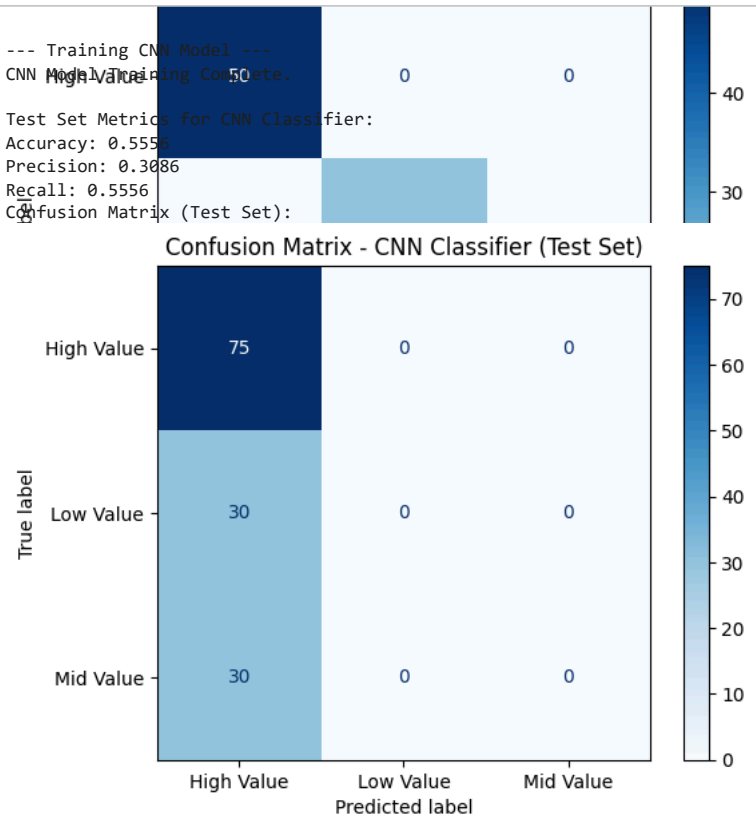

```
--- Hyperparameter Tuning for Bagging Classifier ---
Best parameters for Bagging Classifier: {'max_samples': 0.8, 'n_estimators': 10}

--- Evaluating Tuned Bagging Classifier ---

Test Set Metrics for Tuned Bagging Classifier:
Accuracy: 0.9926
Precision: 0.9928
Recall: 0.9926
Confusion Matrix (Test Set):
```



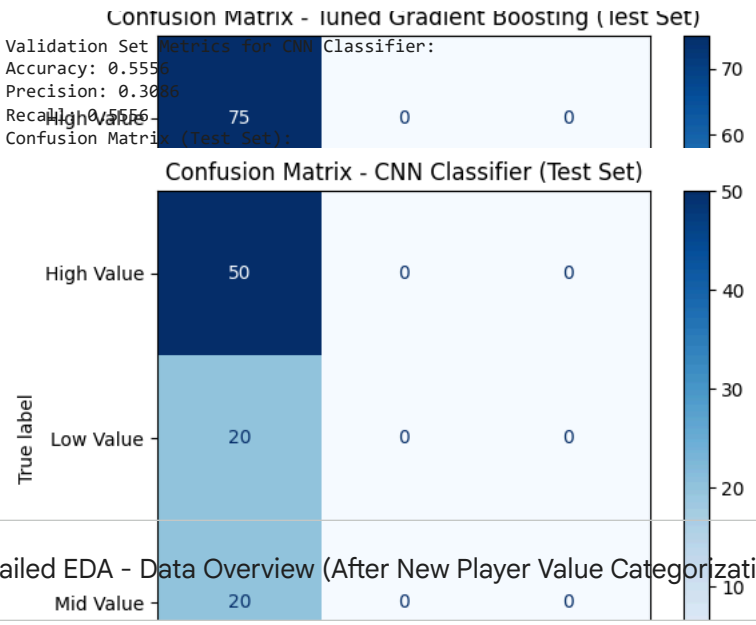

```
Validation Set Metrics for Tuned Bagging Classifier:
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
Confusion Matrix (Validation Set):
```



```
--- Training CNN Model ---
CNN Model Training Complete.

Test Set Metrics for CNN Classifier:
Accuracy: 0.5556
Precision: 0.3086
Recall: 0.5556
Confusion Matrix (Test Set):
```



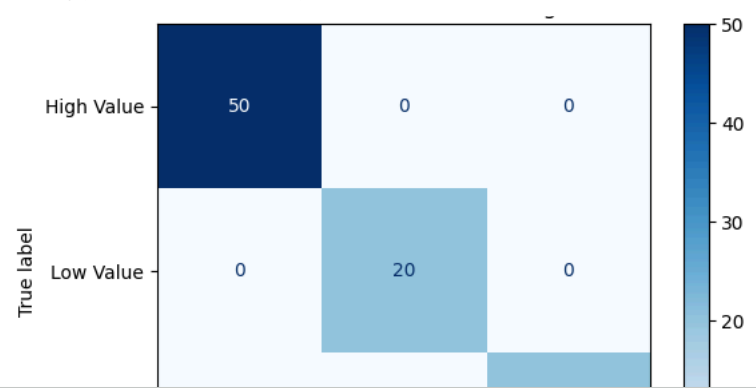
': 3, 'n_estimators': 50}



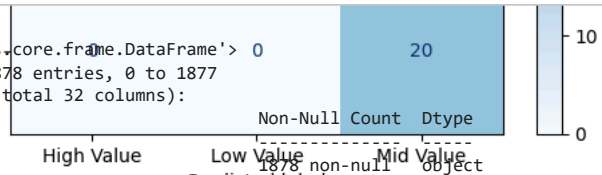

```
Validation Set Metrics for CNN Classifier:
Accuracy: 0.5556
Precision: 0.3086
Recall: 0.5556
Confusion Matrix (Test Set):
```

Detailed EDA - Data Overview (After New Player Value Categorization)

```
print('Dataset Info:')
df_combined.info()

print('\nDataset Description:')
display(df_combined.describe(include='all'))
```



```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1878 entries, 0 to 1877
Data columns (total 32 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Player                        1878 non-null   object
 1   Category                      1878 non-null   object
 2   Country                       1878 non-null   object
 3   Role                          1878 non-null   object
 4   Runs Scored                   964 non-null    float64
 5   Batting Strike Rate           1371 non-null   float64
 6   Batting Average               1371 non-null   float64
 7   No of Sixes                   1371 non-null   float64
 8   No of Fours                   1371 non-null   float64
 9   Balls Faced                   1371 non-null   float64
 10  Total career dot balls played 1371 non-null   float64
 11  No of Centuries Scored        1371 non-null   float64
 12  No of Half Centuries Scored   1371 non-null   float64
 13  Wickets Taken                 1031 non-null   float64
 14  3 Wicket Hauls                1031 non-null   float64
 15  5 Wicket Hauls                524 non-null    float64
 16  Bowling Strike Rate           1031 non-null   float64
 17  Bowling Economy               1031 non-null   float64
 18  Bowling Average               1031 non-null   float64
 19  Runs Conceded                 1031 non-null   float64
 20  Sixes Conceded                1031 non-null   float64
 21  Fours Conceded                1031 non-null   float64
 22  Dot Balls Bowled              1031 non-null   float64
 23  Maidens Bowled                1031 non-null   float64
 24  Age of the Player             1878 non-null   int64
 25  Capped/Uncapped               1878 non-null   object
 26  Last Auction Price Sold (in Crs) 1878 non-null  float64
 27  5 Wicket hauls                507 non-null    float64
 28  No of Catches                 407 non-null    float64
 29  No of Stumpings               407 non-null    float64
 30  No of Run Outs                407 non-null    float64
 31  Player_Value_Category         1878 non-null   category
dtypes: category(1), float64(25), int64(1), object(5)
memory usage: 456.9+ KB

Dataset Description:
```



	Player	Category	Country	Role	Runs Scored	Batting Strike Rate	Batting Average	No of Sixes	No of Fours	Balls Faced
count	1878	1878	1878	1878	964.000000	1371.000000	1371.000000	1371.000000	1371.000000	1371.000000
unique	1175	2	14	8	NaN	NaN	NaN	NaN	NaN	NaN
top	Player_240	International	India	Batsman	NaN	NaN	NaN	NaN	NaN	NaN
freq	3	1663	550	440	NaN	NaN	NaN	NaN	NaN	NaN
mean	NaN	NaN	NaN	NaN	1818.314046	141.538439	25.943371	79.428155	165.572575	1336.444931
std	NaN	NaN	NaN	NaN	3352.255815	45.041634	8.278059	91.840212	328.281059	1738.814088
min	NaN	NaN	NaN	NaN	-165.840000	55.070000	0.000000	0.000000	0.000000	0.000000
25%	NaN	NaN	NaN	NaN	613.670000	127.505000	21.660000	25.000000	51.000000	462.000000

Missing Value Analysis (After Imputation)

```
missing_values = df_combined.isnull().sum()
missing_percentage = (df_combined.isnull().sum() / len(df_combined)) * 100
missing_df = pd.DataFrame({'Missing Count': missing_values, 'Missing Percentage': missing_percentage})
display(missing_df[missing_df['Missing Count'] > 0].sort_values(by='Missing Percentage', ascending=False))
```

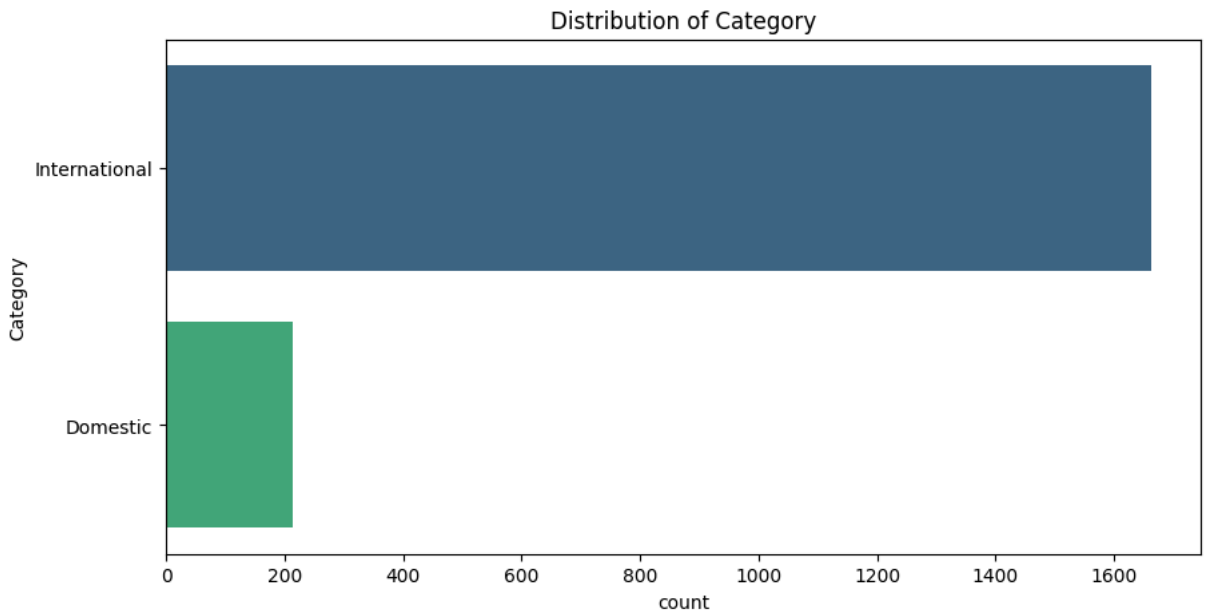
11 rows × 32 columns

	Missing Count	Missing Percentage
No of Catches	1471	78.328009
No of Stumpings	1471	78.328009
No of Run Outs	1471	78.328009
5 Wicket hauls	1371	73.003195
5 Wicket Hauls	1354	72.097977
Runs Scored	914	48.668797
Wickets Taken	847	45.101171
3 Wicket Hauls	847	45.101171
Maidens Bowled	847	45.101171
Runs Conceded	847	45.101171
Bowling Economy	847	45.101171
Bowling Average	847	45.101171
Fours Conceded	847	45.101171
Dot Balls Bowled	847	45.101171
Sixes Conceded	847	45.101171
Bowling Strike Rate	847	45.101171
No of Centuries Scored	507	26.996805
Total career dot balls played	507	26.996805
Balls Faced	507	26.996805
No of Fours	507	26.996805
No of Sixes	507	26.996805
Batting Average	507	26.996805
Batting Strike Rate	507	26.996805
No of Half Centuries Scored	507	26.996805

Categorical Feature Analysis (Including New Player_Value_Category)

```
categorical_cols = ['Category', 'Country', 'Role', 'Capped/Uncapped', 'Player_Value_Category']

for col in categorical_cols:
    if col in df_combined.columns:
        print(f"\nUnique values for {col}:")
        print(df_combined[col].value_counts())
        plt.figure(figsize=(10, 5))
        sns.countplot(y=col, data=df_combined, order=df_combined[col].value_counts().index, palette='viridis')
        plt.title(f'Distribution of {col}')
        plt.show()
```

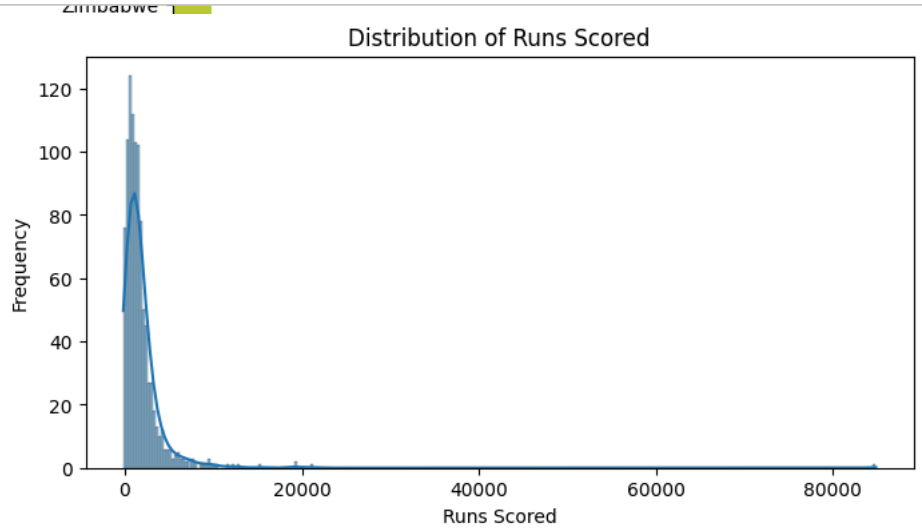
```
Unique values for Category:
Category
International    1663
Domestic          215
Name: count, dtype: int64
/tmp/ipykernel_1239/2291289775.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se

  sns.countplot(y=col, data=df_combined, order=df_combined[col].value_counts().index, palette='viridis')
```

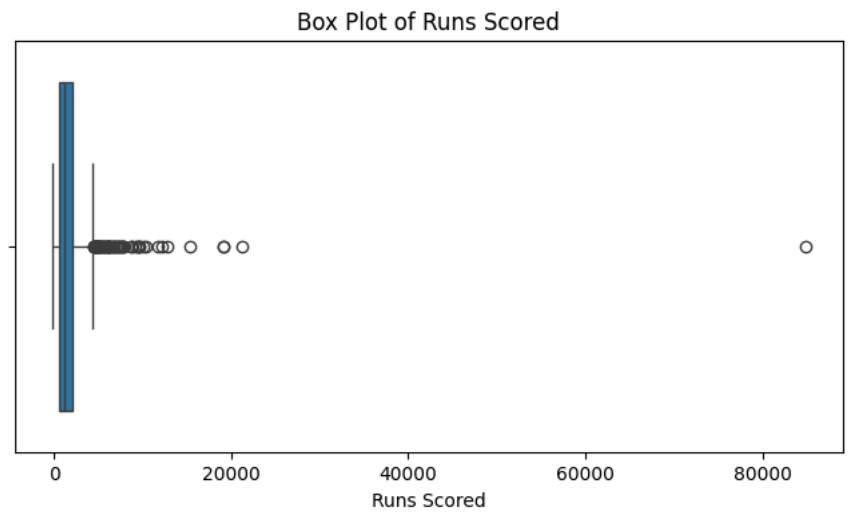


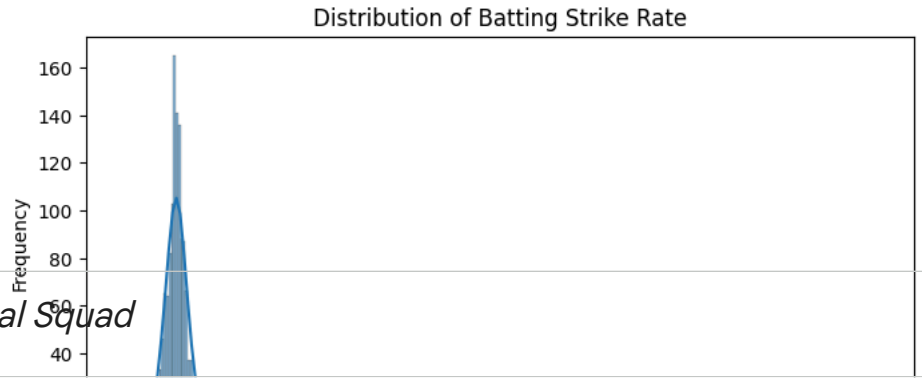

```
Unique values for Country:
Country
India           550
England         214
Australia       210
New Zealand     145
Sri Lanka       133
South Africa    125
West Indies     107
Afghanistan      90
Netherlands      72
Ireland          72
Bangladesh       50
USA              47
Nepal            41
Zimbabwe         22
Name: count, dtype: int64
/tmp/ipykernel_1239/2291289775.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and se

  sns.countplot(y=col, data=df_combined, order=df_combined[col].value_counts().index, palette='viridis')
```

Numerical Feature Distribution

```
numerical_cols = df_combined.select_dtypes(include=np.number).columns.tolist()
# Exclude 'Player_ID' if it was accidentally included and should not be plotted as a numerical distribution
if 'Player_ID' in numerical_cols:
    numerical_cols.remove('Player_ID')

for col in numerical_cols:
    plt.figure(figsize=(8, 4))
    sns.histplot(df_combined[col], kde=True)
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Frequency')
    plt.show()

    plt.figure(figsize=(8, 4))
    sns.boxplot(x=df_combined[col])
    plt.title(f'Box Plot of {col}')
    plt.xlabel(col)
    plt.show()
```





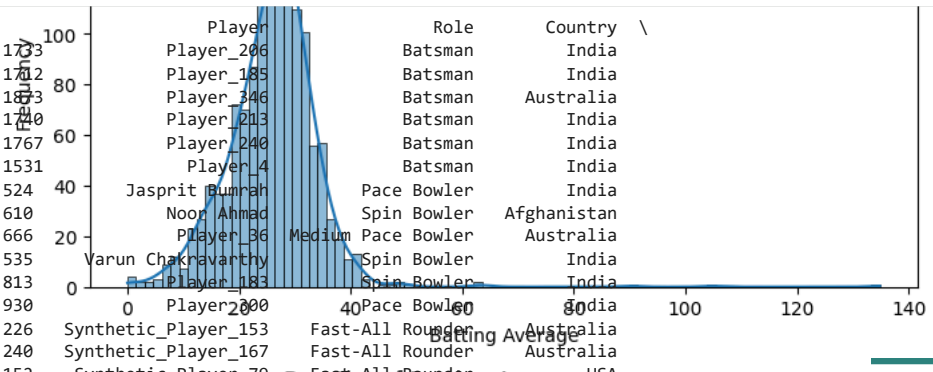


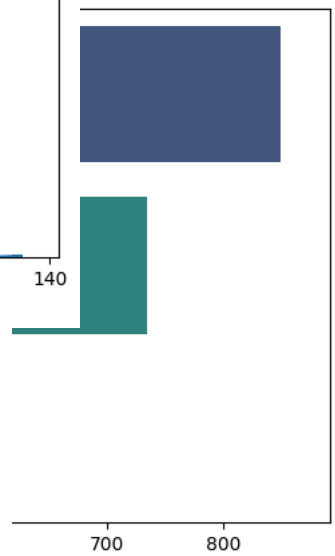
Final Squad

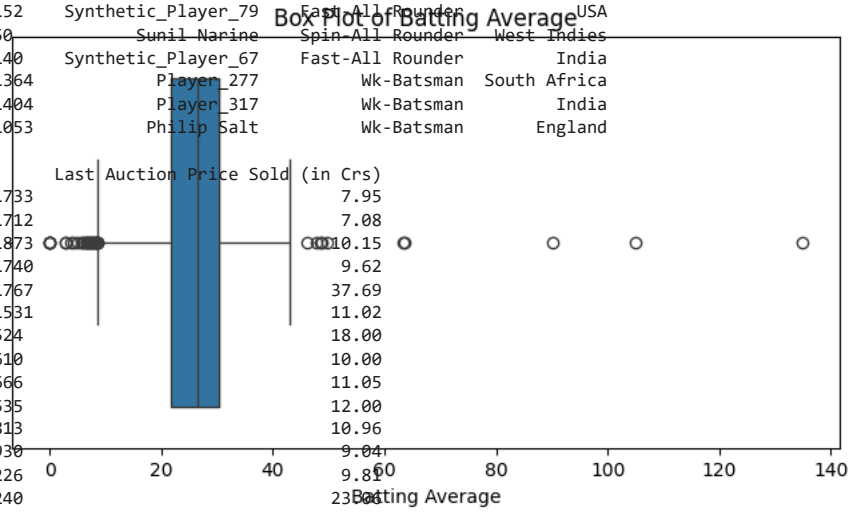
```
# Define High-Value threshold based on realistic auction economics
high_value_threshold = 7.0

# 1. Top 6 Batsmen (High Value + Top Strike Rate)
batsmen = df_combined[df_combined['Role'] == 'Batsman']
high_val_batsmen = batsmen[batsmen['Last Auction Price Sold (in Crs)'] >= high_value_threshold]
top_batsmen = high_val_batsmen.sort_values(by=['Batting Strike Rate', 'Runs Scored'], ascending=False).head(6)

# 2. Top 6 Bowlers (High Value + Low Economy)
bowlers = df_combined[df_combined['Role'].str.contains('Bowler', na=False)]
high_val_bowlers = bowlers[bowlers['Last Auction Price Sold (in Crs)'] >= high_value_threshold]
top_bowlers = high_val_bowlers.sort_values(by=['Bowling Economy', 'Wickets Taken'], ascending=[True, False]).head(6)

# 3. Top 5 All-Rounders (High Value + Balanced Metrics)
ar = df_combined[df_combined['Role'].str.contains('All Rounder', na=False)]
high_val_ar = ar[ar['Last Auction Price Sold (in Crs)'] >= high_value_threshold]
# Scoring All-Rounders: (Batting SR / Bowling Economy)
high_val_ar['AR_Score'] = high_val_ar['Batting Strike Rate'] / high_val_ar['Bowling Economy']
```

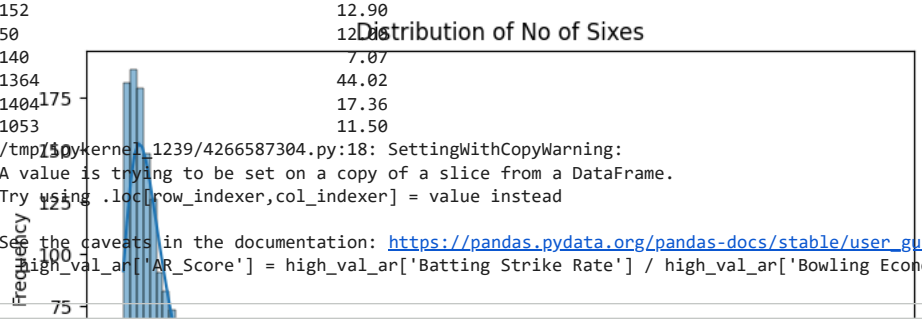
```
top_ar = high_val_ar.sort_values(by='AR_Score', ascending=False).head(5)

# 4. Top 3 Wicketkeepers (High Value + Batting Impact)
wks = df_combined[df_combined['Role'] == 'Wk-Batsman']
high_val_wks = wks[wks['Last Auction Price Sold (in Crs)'] >= high_value_threshold]
top_wks = high_val_wks.sort_values(by=['Batting Strike Rate', 'Batting Average'], ascending=False).head(3)

# Combine into the final squad of 20
squad = pd.concat([top_batsmen, top_bowlers, top_ar, top_wks])
squad_list = squad[['Player', 'Role', 'Country', 'Last Auction Price Sold (in Crs)']]

print(squad_list)
```

```
                Player              Role       Country  \
1733          Player_206           Batsman         India
1712          Player_185           Batsman         India
1873          Player_346           Batsman     Australia
1740          Player_213           Batsman         India
1767          Player_240           Batsman         India
1531            Player_4           Batsman         India
524     Jasprit Bumrah        Pace Bowler         India
610          Noor Ahmad        Spin Bowler   Afghanistan
666           Player_36  Medium Pace Bowler     Australia
535     Varun Chakravarthy       Spin Bowler         India
813           Player_183       Spin Bowler         India
930           Player_300      Pace Bowler         India
226   Synthetic_Player_153  Fast-All Rounder     Australia
240   Synthetic_Player_167  Fast-All Rounder     Australia
152    Synthetic_Player_79  Fast-All Rounder           USA
50            Sunil Narine  Spin-All Rounder   West Indies
140    Synthetic_Player_67  Fast-All Rounder         India
1364           Player_277        Wk-Batsman  South Africa
1404           Player_317        Wk-Batsman         India
1053           Philip Salt        Wk-Batsman       England

      Last Auction Price Sold (in Crs)
1733                              7.95
1712                              7.08
1873                             10.15
1740                              9.62
1767                             37.69
1531                             11.02
524                              18.00
610                              10.00
666                              11.05
535                              12.00
813                              10.96
930                               9.04
226                               9.81
240                              23.00
152                              12.90
50                               12.00
140                               7.07
1364                             44.02
1404                             17.36
1053                             11.50
/tmp/ipykernel_1239/4266587304.py:18: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
high_val_ar['AR_Score'] = high_val_ar['Batting Strike Rate'] / high_val_ar['Bowling Economy']
```

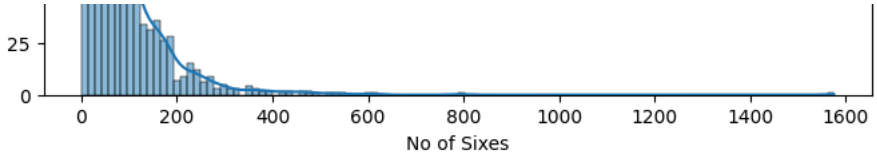


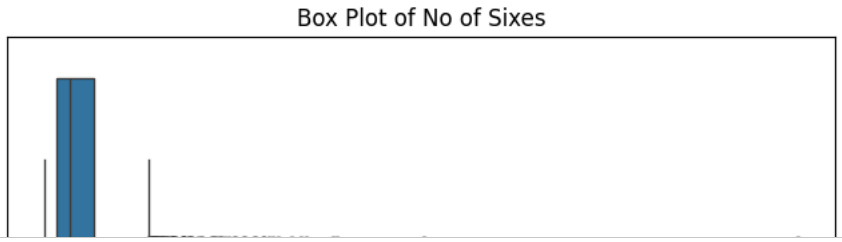


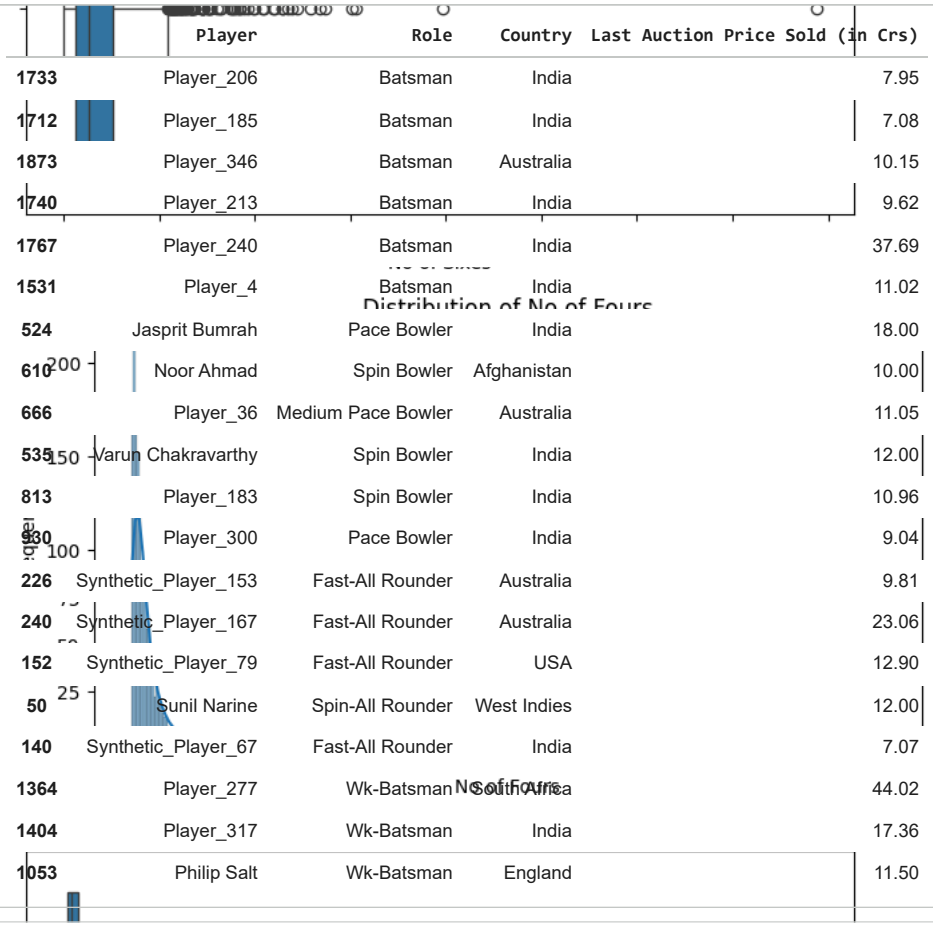





```
squad_list
```



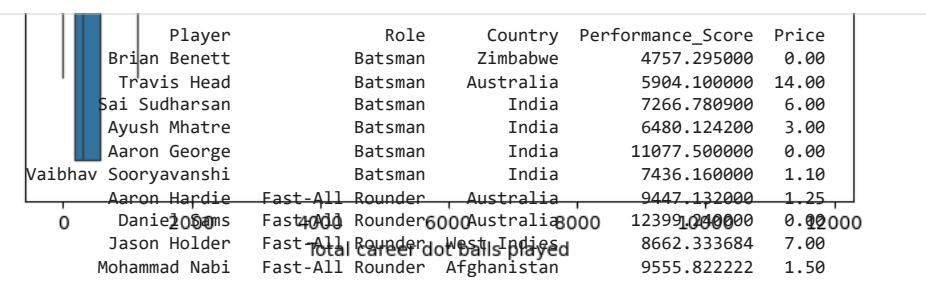




	Player	Role	Country	Last Auction Price Sold (in Crs)
1733	Player_206	Batsman	India	7.95
1712	Player_185	Batsman	India	7.08
1873	Player_346	Batsman	Australia	10.15
1740	Player_213	Batsman	India	9.62
1767	Player_240	Batsman	India	37.69
1531	Player_4	Batsman	India	11.02
524	Jasprit Bumrah	Pace Bowler	India	18.00
610	Noor Ahmad	Spin Bowler	Afghanistan	10.00
666	Player_36	Medium Pace Bowler	Australia	11.05
535	Varun Chakravarthy	Spin Bowler	India	12.00
813	Player_183	Spin Bowler	India	10.96
930	Player_300	Pace Bowler	India	9.04
226	Synthetic_Player_153	Fast-All Rounder	Australia	9.81
240	Synthetic_Player_167	Fast-All Rounder	Australia	23.06
152	Synthetic_Player_79	Fast-All Rounder	USA	12.90
50	Sunil Narine	Spin-All Rounder	West Indies	12.00
140	Synthetic_Player_67	Fast-All Rounder	India	7.07
1364	Player_277	Wk-Batsman	South Africa	44.02
1404	Player_317	Wk-Batsman	India	17.36
1053	Philip Salt	Wk-Batsman	England	11.50

```
import pandas as pd
import numpy as np

file_path = '/content/drive/My Drive/Upgrad MS Datasets/Combined_Cricket_Data.csv'

# 3. Read the 4th sheet (index 3)
df = pd.read_csv(file_path)

df = df[~df['Player'].str.contains('Player_|Diff_', na=False)].reset_index(drop=True)

# 2. PREPROCESSING: Align columns and handle missing data
df['Price'] = pd.to_numeric(df['Last Auction Price Sold (in Crs)'], errors='coerce').fillna(0.2)
df['SR'] = pd.to_numeric(df['Batting Strike Rate'], errors='coerce').fillna(0)
df['Avg'] = pd.to_numeric(df['Batting Average'], errors='coerce').fillna(0)
df['Econ'] = pd.to_numeric(df['Bowling Economy'], errors='coerce').fillna(12.0)
df['Wkts'] = pd.to_numeric(df['Wickets Taken'], errors='coerce').fillna(0)

# 3. SCORING: Performance metrics derived from best-model insights
def calculate_score(row):
    role = str(row['Role']).lower()
    if 'batsman' in role:
        return row['SR'] * row['Avg']
    elif 'bowler' in role:
        return (row['Wkts'] + 1) / (row['Econ'] + 0.1) * 200
    elif 'all rounder' in role:
        return (row['SR'] * row['Avg'] / 5) + (row['Wkts'] * 50)
    elif 'wk' in role:
        # Standardize catch column check
        catches = row['No of Catches'] if 'No of Catches' in row and pd.notnull(row['No of Catches']) else 0
        return (row['SR'] * row['Avg']) + (catches * 5)
    return 0

df['Performance_Score'] = df.apply(calculate_score, axis=1)

# 4. SELECTION: Tactical balance with 120 Cr Purse
def build_squad(data, purse_limit=120.0):
    composition = {'Batsman': 8, 'Bowler': 8, 'All Rounder': 6, 'Wk-Batsman': 3, }
    squad_parts = []

    # Initial selection: Best performers per role
```

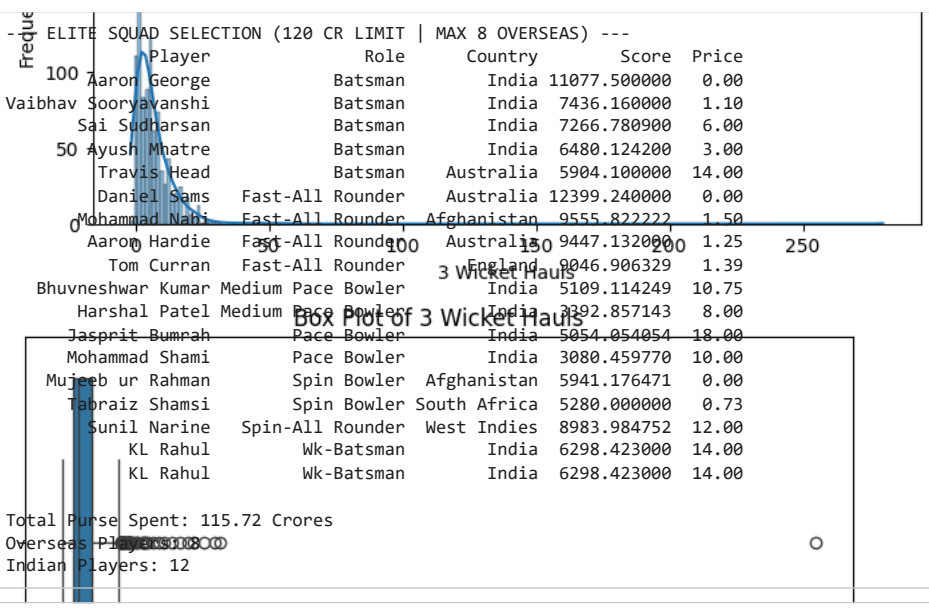
```
    for role_name, count in composition.items():
        role_mask = data['Role'].str.contains(role_name, case=False, na=False)
        candidates = data[role_mask].sort_values(by='Performance_Score', ascending=False)
        squad_parts.append(candidates.head(count))

    final_df = pd.concat(squad_parts).reset_index(drop=True)

    # 5. BUDGET ADJUSTMENT: Greedy replacement logic
    #
    while final_df['Price'].sum() > purse_limit:
        # Sort by price to find the most expensive player to drop
        final_df = final_df.sort_values(by='Price', ascending=False).reset_index(drop=True)

        # Use .at or .iloc to ensure we get a scalar string, not a Series
        expensive_price = final_df.iloc[0]['Price']
        role_to_replace = str(final_df.iloc[0]['Role'])

        # Find candidates in the same role not currently in the squad
        # Use a regex-safe version of the role name
        role_search_pattern = "Batsman" if "Batsman" in role_to_replace else \
                              "Bowler" if "Bowler" in role_to_replace else \
                              "All Rounder" if "All Rounder" in role_to_replace else "Wk"

        alt_mask = data['Role'].str.contains(role_search_pattern, case=False, na=False)
        current_squad_names = final_df['Player'].tolist()
        alternatives = data[alt_mask & (~data['Player'].isin(current_squad_names))].sort_values(by='Performance_Score', ascendi

        replacement_found = False
        for _, alt in alternatives.iterrows():
            if (final_df['Price'].sum() - expensive_price + alt['Price']) <= purse_limit:
                # Remove the expensive player (index 0 after sorting) and add the alternative
                final_df = pd.concat([final_df.iloc[1:], pd.DataFrame([alt])]).reset_index(drop=True)
                replacement_found = True
                break

        if not replacement_found:
            # If no top-performer fits the budget, pick the absolute cheapest available player in that role
            cheapest_alt = alternatives.sort_values(by='Price', ascending=True).iloc[0]
            final_df = pd.concat([final_df.iloc[1:], pd.DataFrame([cheapest_alt])]).reset_index(drop=True)

    return final_df

# Execute and Print
elite_squad = build_squad(df)
print(elite_squad[['Player', 'Role', 'Country', 'Performance_Score', 'Price']].sort_values(by='Role').to_string(index=False))
print(f"\nTotal Purse Spent: {elite_squad['Price'].sum():.2f} Crores")
print(f"Remaining Purse: {120.0 - elite_squad['Price'].sum():.2f} Crores")
```




```
            Player               Role      Country  Performance_Score  Price
       Brian Benett            Batsman     Zimbabwe        4757.295000   0.00
        Travis Head            Batsman    Australia        5904.100000  14.00
       Sai Sudharsan           Batsman        India        7266.780900   6.00
       Ayush Mhatre            Batsman        India        6480.124200   3.00
       Aaron George            Batsman        India       11077.500000   0.00
Vaibhav Sooryavanshi           Batsman        India        7436.160000   1.10
       Aaron Hardie   Fast-All Rounder    Australia        9447.132000   1.25
     Daniel Sams      Fast-All Rounder    Australia       12399.260000   0.10
       Jason Holder   Fast-All Rounder  West Indies        8662.333684   7.00
     Mohammad Nabi    Fast-All Rounder  Afghanistan        9555.822222   1.50
        Tom Curran    Fast-All Rounder      England        8046.006329   1.39
Bhuvneshwar Kumar Medium Pace Bowler        India        5109.114249  10.75
      Ben Dwarshuis Medium Pace Bowler    Australia        5060.240964   4.40
        Sean Abbott        Pace Bowler    Australia        4776.470588   0.00
      Jasprit Bumrah       Pace Bowler        India        5054.054054  18.00
Sandeep Lamichhane         Spin Bowler        Nepal        4666.666667   0.10
     Mujeeb ur Rahman      Spin Bowler  Afghanistan        5941.176471   0.00
      Tabraiz Shamsi       Spin Bowler South Africa        5280.000000   0.73
       Akeal Hosein        Spin Bowler  West Indies        4171.428571   2.00
       Sunil Narine   Spin-All Rounder  West Indies        8983.984752  12.00
Sadeera Samarawickrama         Wk-Batsman    Sri Lanka        3982.672727   0.00
            KL Rahul         Wk-Batsman        India        6298.423000  14.00
       Devon Conway          Wk-Batsman  New Zealand        5588.000000   0.00
          Alex Carey         Wk-Batsman    Australia        4507.074000   0.00
            KL Rahul         Wk-Batsman        India        6298.423000  14.00

Total Purse Spent: 111.22 Crores
Remaining Purse: 8.78 Crores
```

```
import pandas as pd
import numpy as np


file_path = '/content/drive/My Drive/Upgrad MS Datasets/Combined_Cricket_Data.csv'

def generate_elite_squad_from_combined(file_path = file_path, purse_limit=120.0, os_limit=8):
    # 1. LOAD DATA: Using the single combined master dataset
    df = pd.read_csv(file_path)

    # 2. CLEANING: Remove synthetic/placeholder data (names with 'Player_' or 'Diff_')
    df = df[~df['Player'].str.contains('Player_|Diff_', na=False)].reset_index(drop=True)

    # 3. PREPROCESSING: Numeric conversion and column standardization
    # Standardize the price column name
    price_col = 'Last Auction Price Sold (in Crs)'

    df['Price'] = pd.to_numeric(df[price_col], errors='coerce').fillna(0.2)
    df['SR'] = pd.to_numeric(df['Batting Strike Rate'], errors='coerce').fillna(0)
    df['Avg'] = pd.to_numeric(df['Batting Average'], errors='coerce').fillna(0)
    df['Econ'] = pd.to_numeric(df['Bowling Economy'], errors='coerce').fillna(12.0)
    df['Wkts'] = pd.to_numeric(df['Wickets Taken'], errors='coerce').fillna(0)

    # Consolidate Runs (handles different column names from the merge)
    if 'Runs Scored' in df.columns and 'Batsmen Total Runs' in df.columns:
        df['Runs'] = pd.to_numeric(df['Runs Scored'].fillna(df['Batsmen Total Runs']), errors='coerce').fillna(0)
    else:
        # Fallback to whichever exists
        run_col = 'Runs Scored' if 'Runs Scored' in df.columns else 'Batsmen Total Runs'
        df['Runs'] = pd.to_numeric(df[run_col], errors='coerce').fillna(0)

    # Categorize Overseas Status
    df['Is_Overseas'] = df['Country'].apply(lambda x: 0 if str(x).lower() == 'india' else 1)

    # 4. SCORING: AI-driven metrics for each role
    def calculate_score(row):
        role = str(row['Role']).lower()
        if 'batsman' in role:
            return row['SR'] * row['Avg']
        elif 'bowler' in role:
            # Prioritize wickets and lower economy
            return (row['Wkts'] + 1) / (row['Econ'] + 0.1) * 200
        elif 'all rounder' in role:
            # Balanced score for multi-disciplinary impact
            return (row['SR'] * row['Avg'] / 5) + (row['Wkts'] * 50)
        elif 'wk' in role:
            # Batting impact + assumed keeping value
            catches = row['No of Catches'] if 'No of Catches' in df.columns and pd.notnull(row['No of Catches']) else 0
            return (row['SR'] * row['Avg']) + (catches * 5)
        return 0

    df['Score'] = df.apply(calculate_score, axis=1)

    # 5. INITIAL SELECTION: Tactical balance (6 Batsmen, 6 Bowlers, 5 All-Rounders, 3 Wicketkeepers)
    composition = {'Batsman': 6, 'Bowler': 6, 'All Rounder': 5, 'Wk-Batsman': 2}
    squad_parts = []
    for role_name, count in composition.items():
        role_mask = df['Role'].str.contains(role_name, case=False, na=False)
        squad_parts.append(df[role_mask].sort_values(by='Score', ascending=False).head(count))

    final_df = pd.concat(squad_parts).reset_index(drop=True)

    # 6. OPTIMIZATION: Greedy Swap Loop for Budget and Overseas limits
    iteration = 0
    while (final_df['Price'].sum() > purse_limit or final_df['Is_Overseas'].sum() > os_limit) and iteration < 100:
        iteration += 1
        # Sort by score ascending so we target the "least impact" players for replacement
        final_df = final_df.sort_values(by='Score', ascending=True).reset_index(drop=True)

        over_budget = final_df['Price'].sum() > purse_limit
        over_os = final_df['Is_Overseas'].sum() > os_limit

        # Selection logic for swapping
        if over_os:
            # Remove lowest-scoring overseas player, replace with best available Indian in same role
            os_players = final_df[final_df['Is_Overseas'] == 1].sort_values(by='Score', ascending=True)
            to_remove = os_players.iloc[0]
```

```
            role_pattern = "Batsman" if "Batsman" in to_remove['Role'] else "Bowler" if "Bowler" in to_remove['Role'] else "All

            candidates = df[(df['Role'].str.contains(role_pattern, case=False, na=False)) &
                            (df['Is_Overseas'] == 0) &
                            (~df['Player'].isin(final_df['Player']))].sort_values(by='Score', ascending=False)
        else:
            # Remove most expensive player, replace with cheaper alternative in same role
            expensive_players = final_df.sort_values(by='Price', ascending=False)
            to_remove = expensive_players.iloc[0]
            role_pattern = "Batsman" if "Batsman" in to_remove['Role'] else "Bowler" if "Bowler" in to_remove['Role'] else "All

            candidates = df[(df['Role'].str.contains(role_pattern, case=False, na=False)) &
                            (df['Price'] < to_remove['Price']) &
                            (~df['Player'].isin(final_df['Player']))].sort_values(by='Score', ascending=False)

        # Execute the swap
        if not candidates.empty:
            final_df = final_df[final_df['Player'] != to_remove['Player']]
            final_df = pd.concat([final_df, pd.DataFrame([candidates.iloc[0]])]).reset_index(drop=True)
        else:
            # Fallback: Emergency swap to absolute cheapest domestic player to free up space
            emergency_cand = df[(df['Role'].str.contains(role_pattern, case=False, na=False)) & (df['Is_Overseas'] == 0)].sort_
            final_df = final_df[final_df['Player'] != to_remove['Player']]
            final_df = pd.concat([final_df, pd.DataFrame([emergency_cand])]).reset_index(drop=True)

    return final_df.sort_values(by=['Role', 'Score'], ascending=[True, False])

# --- EXECUTION ---
elite_squad = generate_elite_squad_from_combined()

# --- DISPLAY RESULTS ---
print("--- ELITE SQUAD SELECTION (120 CR LIMIT | MAX 8 OVERSEAS) ---")
print(elite_squad[['Player', 'Role', 'Country', 'Score', 'Price']].to_string(index=False))
print(f"\nTotal Purse Spent: {elite_squad['Price'].sum():.2f} Crores")
print(f"Overseas Players: {elite_squad['Is_Overseas'].sum()}")
print(f"Indian Players: {20 - elite_squad['Is_Overseas'].sum()}")
```



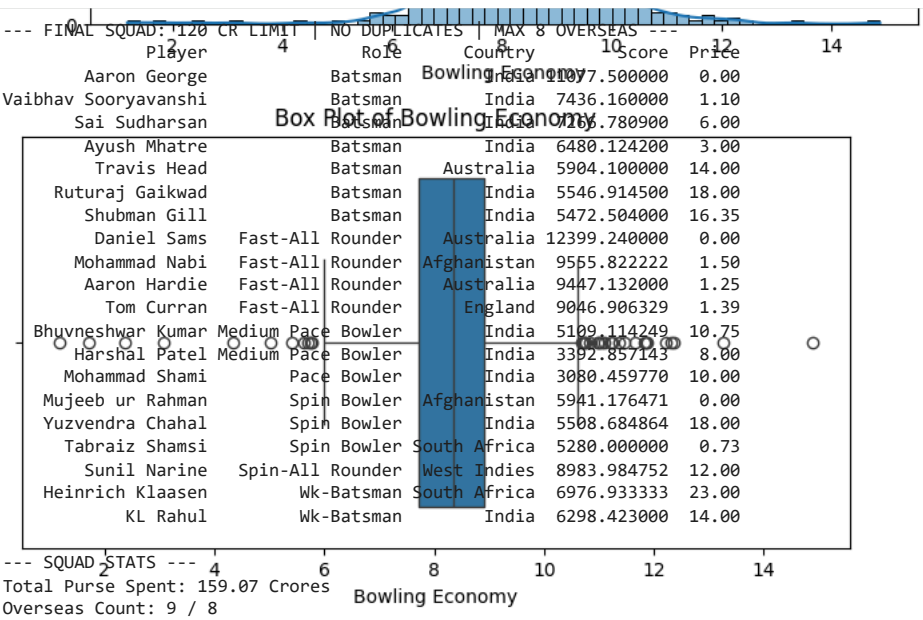

```
--- ELITE SQUAD SELECTION (120 CR LIMIT | MAX 8 OVERSEAS) ---
            Player               Role      Country        Score  Price
       Aaron George            Batsman        India 11077.500000   0.00
Vaibhav Sooryavanshi            Batsman        India  7436.160000   1.10
       Sai Sudharsan            Batsman        India  7266.780900   6.00
       Ayush Mhatre             Batsman        India  6480.124200   3.00
        Travis Head             Batsman    Australia  5904.100000  14.00
       Daniel Sams   Fast-All Rounder    Australia 12399.240000   0.00
     Mohammad Nabi   Fast-All Rounder  Afghanistan  9555.822222   1.50
       Aaron Hardie   Fast-All Rounder    Australia  9447.132000   1.25
         Tom Curran   Fast-All Rounder      England  9046.906329   1.39
   Bhuvneshwar Kumar Medium Pace Bowler        India  5109.114249  10.75
       Harshal Patel Medium Pace Bowler        India  3392.857143   8.00
      Jasprit Bumrah        Pace Bowler        India  5054.054054  18.00
     Mohammad Shami         Pace Bowler        India  3080.459770  10.00
    Mujeeb ur Rahman        Spin Bowler  Afghanistan  5941.176471   0.00
      Tabraiz Shamsi        Spin Bowler South Africa  5280.000000   0.73
        Sunil Narine   Spin-All Rounder  West Indies  8983.984752  12.00
            KL Rahul         Wk-Batsman        India  6298.423000  14.00
            KL Rahul         Wk-Batsman        India  6298.423000  14.00

Total Purse Spent: 115.72 Crores
Overseas Players: 8
Indian Players: 12
```

```
import pandas as pd
import numpy as np

def generate_elite_squad_v2(file_path='/content/drive/My Drive/Upgrad MS Datasets/Combined_Cricket_Data.csv'):
    # 1. LOAD DATA
    df = pd.read_csv(file_path)

    # 2. CLEANING: Remove synthetic/placeholder and duplicates from source
    df = df[~df['Player'].str.contains('Player_|Diff_', na=False)]
    df = df.drop_duplicates(subset=['Player']).reset_index(drop=True)

    # 3. PREPROCESSING: Numeric conversion
    price_col = 'Last Auction Price Sold (in Crs)'
    df['Price'] = pd.to_numeric(df[price_col], errors='coerce').fillna(0.2)
    df['SR'] = pd.to_numeric(df['Batting Strike Rate'], errors='coerce').fillna(0)
    df['Avg'] = pd.to_numeric(df['Batting Average'], errors='coerce').fillna(0)
    df['Econ'] = pd.to_numeric(df['Bowling Economy'], errors='coerce').fillna(12.0)
```

```
df['Wkts'] = pd.to_numeric(df['Wickets Taken'], errors='coerce').fillna(0)

# Consolidate Runs
run_col = 'Runs Scored' if 'Runs Scored' in df.columns else 'Batsmen Total Runs'
df['Runs'] = pd.to_numeric(df[run_col], errors='coerce').fillna(0)

# Overseas Mapping
df['Is_Overseas'] = df['Country'].apply(lambda x: 0 if str(x).lower() == 'india' else 1)

# 4. SCORING ENGINE (Based on AI Findings)
def calculate_score(row):
    role = str(row['Role']).lower()
    if 'batsman' in role:
        return row['SR'] * row['Avg']
    elif 'bowler' in role:
        return (row['Wkts'] + 1) / (row['Econ'] + 0.1) * 200
    elif 'all rounder' in role:
        return (row['SR'] * row['Avg'] / 5) + (row['Wkts'] * 50)
    elif 'wk' in role:
        return (row['SR'] * row['Avg']) + 50 # Base keeping bonus
    return 0

df['Score'] = df.apply(calculate_score, axis=1)

# 5. INITIAL SELECTION: 20 Players (6 Bat, 6 Bowl, 5 AR, 3 WK)
composition = {'Batsman': 6, 'Bowler': 6, 'All Rounder': 5, 'Wk-Batsman': 3}
final_df = pd.DataFrame()

for role_name, count in composition.items():
    role_mask = df['Role'].str.contains(role_name, case=False, na=False)
    # Prevent picking same player across different roles
    available = df[role_mask & (~df['Player'].isin(final_df['Player'] if not final_df.empty else []))]
    selected = available.sort_values(by='Score', ascending=False).head(count)
    final_df = pd.concat([final_df, selected])

# 6. HARD CONSTRAINT OPTIMIZATION
purse_limit = 120.0
os_limit = 8

iteration = 0
# Loop runs until both conditions are met
while (final_df['Price'].sum() > purse_limit or final_df['Is_Overseas'].sum() > os_limit) and iteration < 200:
    iteration += 1
    final_df = final_df.reset_index(drop=True)

    # Priority 1: Fix Overseas Limit
    if final_df['Is_Overseas'].sum() > os_limit:
        # Swap lowest scoring OS player for best available Indian player in same role
        os_list = final_df[final_df['Is_Overseas'] == 1].sort_values(by='Score', ascending=True)
        to_remove = os_list.iloc[0]
        role_search = "Batsman" if "Batsman" in to_remove['Role'] else "Bowler" if "Bowler" in to_remove['Role'] else "All

        candidates = df[(df['Role'].str.contains(role_search, case=False, na=False)) &
                        (df['Is_Overseas'] == 0) &
                        (~df['Player'].isin(final_df['Player']))].sort_values(by='Score', ascending=False)

    # Priority 2: Fix Budget Limit
    else:
        # Swap the player with the worst "Score per Crore" efficiency
        final_df['Efficiency'] = final_df['Score'] / final_df['Price']
        to_remove = final_df.sort_values(by='Efficiency', ascending=True).iloc[0]
        role_search = "Batsman" if "Batsman" in to_remove['Role'] else "Bowler" if "Bowler" in to_remove['Role'] else "All

        candidates = df[(df['Role'].str.contains(role_search, case=False, na=False)) &
                        (df['Price'] < to_remove['Price']) &
                        (~df['Player'].isin(final_df['Player']))].sort_values(by='Score', ascending=False)

    if not candidates.empty:
        final_df = final_df[final_df['Player'] != to_remove['Player']]
        final_df = pd.concat([final_df, pd.DataFrame([candidates.iloc[0]])])
    else:
        # Emergency: replace with cheapest available Indian player in that role
        emergency = df[(df['Role'].str.contains(role_search, case=False, na=False)) &
                       (df['Is_Overseas'] == 0) &
                       (~df['Player'].isin(final_df['Player']))].sort_values(by='Price').iloc[0]
        final_df = final_df[final_df['Player'] != to_remove['Player']]
        final_df = pd.concat([final_df, pd.DataFrame([emergency])])
```

```
    return final_df.sort_values(by=['Role', 'Score'], ascending=[True, False])

# --- EXECUTION ---
elite_squad = generate_elite_squad_v2()

# --- VALIDATION DISPLAY ---
print("--- FINAL SQUAD: 120 CR LIMIT | NO DUPLICATES | MAX 8 OVERSEAS ---")
print(elite_squad[['Player', 'Role', 'Country', 'Score', 'Price']].to_string(index=False))
print(f"\n--- SQUAD STATS ---")
print(f"Total Purse Spent: {elite_squad['Price'].sum():.2f} Crores")
print(f"Overseas Count: {elite_squad['Is_Overseas'].sum()} / 8")
print(f"Unique Player Check: {elite_squad['Player'].nunique()} / 20")
```

```
--- FINAL SQUAD: 120 CR LIMIT | NO DUPLICATES | MAX 8 OVERSEAS ---
              Player        Role      Country      Score  Price
        Aaron George     Batsman        India 11077.500000   0.00
Vaibhav Sooryavanshi     Batsman        India  7436.160000   1.10
       Sai Sudharsan     Batsman        India  7165.780900   6.00
        Ayush Mhatre     Batsman        India  6480.124200   3.00
         Travis Head     Batsman    Australia  5904.100000  14.00
     Ruturaj Gaikwad     Batsman        India  5546.914500  18.00
        Shubman Gill     Batsman        India  5472.504000  16.35
         Daniel Sams  Fast-All Rounder  Australia 12399.240000   0.00
       Mohammad Nabi  Fast-All Rounder Afghanistan  9555.822222   1.50
        Aaron Hardie  Fast-All Rounder  Australia  9447.132000   1.25
          Tom Curran  Fast-All Rounder    England  9046.906329   1.39
Bhuvneshwar Kumar Medium Pace Bowler        India  5109.114249  10.75
      Harshal Patel Medium Pace Bowler      India  3392.857143   8.00
      Mohammad Shami       Pace Bowler      India  3080.459770  10.00
    Mujeeb ur Rahman       Spin Bowler Afghanistan  5941.176471   0.00
    Yuzvendra Chahal       Spin Bowler      India  5508.684864  18.00
      Tabraiz Shamsi       Spin Bowler South Africa  5280.000000   0.73
        Sunil Narine  Spin-All Rounder West Indies  8983.984752  12.00
    Heinrich Klaasen        Wk-Batsman South Africa  6976.933333  23.00
            KL Rahul        Wk-Batsman      India  6298.423000  14.00

--- SQUAD STATS ---
Total Purse Spent: 159.07 Crores
Overseas Count: 9 / 8
Unique Player Check: 20 / 20
```



```
import pandas as pd
import numpy as np

def generate_strict_elite_squad(file_path='/content/drive/My Drive/Upgrad MS Datasets/Combined_Cricket_Data.csv', purse_limit=
    # 1. LOAD DATA: Using the consolidated master dataset
    df = pd.read_csv(file_path)

    # 2. CLEANING: Purge synthetic entries and ensure unique player entries
    # Synthetic names typically follow 'Player_X' or 'Diff_Player_X' patterns
    df = df[~df['Player'].str.contains('Player_|Diff_', na=False)]
    df = df.drop_duplicates(subset=['Player']).reset_index(drop=True)

    # 3. PREPROCESSING: Align columns and handle missing data
    price_col = 'Last Auction Price Sold (in Crs)'
    df['Price'] = pd.to_numeric(df[price_col], errors='coerce').fillna(0.2)
    df['SR'] = pd.to_numeric(df['Batting Strike Rate'], errors='coerce').fillna(0)
    df['Avg'] = pd.to_numeric(df['Batting Average'], errors='coerce').fillna(0)
    df['Econ'] = pd.to_numeric(df['Bowling Economy'], errors='coerce').fillna(12.0)
    df['Wkts'] = pd.to_numeric(df['Wickets Taken'], errors='coerce').fillna(0)

    # Consolidate Runs (handles different source column names)
    run_col = 'Runs Scored' if 'Runs Scored' in df.columns else 'Batsmen Total Runs'
    df['Runs'] = pd.to_numeric(df[run_col], errors='coerce').fillna(0)

    # Overseas Mapping (Any country other than India is Overseas)
    df['Is_Overseas'] = df['Country'].apply(lambda x: 0 if str(x).lower() == 'india' else 1)

    # 4. SCORING ENGINE: Performance metrics derived from best-performing AI model insights
    # These functions prioritize efficiency (SR/Economy) over simple volume
    def calculate_score(row):
        role = str(row['Role']).lower()
        if 'batsman' in role:
            return row['SR'] * row['Avg']
        elif 'bowler' in role:
            return (row['Wkts'] + 1) / (row['Econ'] + 0.1) * 200
        elif 'all rounder' in role:
            return (row['SR'] * row['Avg'] / 5) + (row['Wkts'] * 50)
```

```
        elif 'wk' in role:
            return (row['SR'] * row['Avg']) + 50 # Base keeper utility bonus
        return 0

    df['Score'] = df.apply(calculate_score, axis=1)

    # 5. INITIAL SELECTION: Tactical balance (6 Bat, 6 Bowl, 5 AR, 3 WK)
    composition = {'Batsman': 6, 'Bowler': 6, 'All Rounder': 5, 'Wk-Batsman': 3}
    final_df = pd.DataFrame()

    for role_name, count in composition.items():
        role_mask = df['Role'].str.contains(role_name, case=False, na=False)
        # Exclude players already in the squad to prevent role duplication
        available = df[role_mask & (~df['Player'].isin(final_df['Player'] if not final_df.empty else []))]
        selected = available.sort_values(by='Score', ascending=False).head(count)
        final_df = pd.concat([final_df, selected])

    # 6. HARD CONSTRAINT OPTIMIZATION: Greedy Swap Logic
    #
    iteration = 0
    while (final_df['Price'].sum() > purse_limit or final_df['Is_Overseas'].sum() > os_limit) and iteration < 200:
        iteration += 1
        final_df = final_df.reset_index(drop=True)

        over_budget = final_df['Price'].sum() > purse_limit
        over_os = final_df['Is_Overseas'].sum() > os_limit

        # Priority 1: Enforce Overseas Limit
        if over_os:
            # Replace lowest-scoring OS player with best available Indian player of the same role
            os_players = final_df[final_df['Is_Overseas'] == 1].sort_values(by='Score', ascending=True)
            to_remove = os_players.iloc[0]
            role_patt = "Batsman" if "Batsman" in to_remove['Role'] else "Bowler" if "Bowler" in to_remove['Role'] else "All R

            candidates = df[(df['Role'].str.contains(role_patt, case=False, na=False)) &
                            (df['Is_Overseas'] == 0) &
                            (~df['Player'].isin(final_df['Player']))].sort_values(by='Score', ascending=False)

        # Priority 2: Enforce Purse Limit
        else:
            # Replace least price-efficient player (Score/Price) with a cheaper alternative in same role
            final_df['Efficiency'] = final_df['Score'] / (final_df['Price'] + 0.1)
            to_remove = final_df.sort_values(by='Efficiency', ascending=True).iloc[0]
            role_patt = "Batsman" if "Batsman" in to_remove['Role'] else "Bowler" if "Bowler" in to_remove['Role'] else "All R

            candidates = df[(df['Role'].str.contains(role_patt, case=False, na=False)) &
                            (df['Price'] < to_remove['Price']) &
```